

## EXPERIENCE

2026.1 – Present **Sr. AI Engineer**, Selta Square**AI Medical Assessment: LLM multi-agent ICSR causality assessment** 2026.1 – Present

Main AI engineer and PM of a role-separated multi-agent system that drafts WHO-UMC causality assessments with line-level evidence for assessors to finalize; designed 24 assessor-preference profiles and a three-axis evaluation, through production and an enterprise beta.

**In-house pharmacovigilance assistant agent and tools** 2026.4 – Present

In-house PV assistant agent that calls regulatory, literature and MedDRA coding tools along the PV workflow and confirms each judgment with the user; built its tools, workflow and gates.

**ICSR case processing: E2B(R3) extraction and auto entry** 2026.1 – Present

E2B(R3) extraction and auto entry that separates form labels from entered content and feeds them hierarchically, since SAE/CIOMS reports are mostly fixed forms; with source tracing and a review UI.

2024.8 – 2025.12 **Research Assistant Professor**, Biomedical Research Institute, Seoul National University Hospital2023.3 – 2024.7 **Post-doctoral Researcher**, Seoul National University Hospital**Zero-shot NER via disagreement signals across multiple LLMs** 2023.3 – 2025.10

Label-free method that simulates pilot annotation, refining annotation instructions from disagreements among heterogeneous LLMs; reached zero-shot state of the art without instruction fine-tuning (ACL 2026 Main).

**How well medical QA benchmarks reflect the clinical capabilities of LLMs** 2025.2 – 2025.8

Systematically correlated LLM scores on major medical QA benchmarks (MedQA, MedMCQA, PubMedQA, MMLU medical subjects) with real-world clinical performance; benchmarks only partially reflect clinical capability.

**In-context learning for clinical information extraction from hematology-oncology notes** 2023.3 – 2025.10

Extracting diagnoses, symptoms, and prescriptions using few-shot retrieval and LLM in-context learning.

2018.2 – 2023.2 **Integrated M.S./Ph.D. Researcher**, Seoul National University**Revisiting ML-based drug repositioning with clinical-trial eligibility criteria** Microsoft Research Asia · 2022.5 – 2023.4

· Project lead

Showed that drug indications are not a sound prediction target for ML-based drug repositioning (CHIL 2023), and reframed the problem as recommending clinical-trial eligibility criteria with a public benchmark and an automatic evaluation framework (Findings of EACL 2024).

**NLP for extracting drug safety information from adverse event reports** Ministry of Food and Drug Safety · 2021.2 – 2022.2

· Project lead

KAERS-BERT for mixed Korean-English narratives; ICH E2B(R3)-based annotated corpus; NER, RE, sentence extraction and label classification models.

## EDUCATION

2018.3 – 2023.2 **Ph.D. in Medicine (Applied Bio-Engineering)**, Seoul National University

Thesis: *Clinical Information Extraction from Unstructured Free-texts for Pharmacovigilance* · Advisor: Howard Lee

2013.3 – 2018.2 **B.S. in Biology and Anthropology**, Seoul National University — Summa Cum Laude

## SELECTED PUBLICATIONS

1. **DiZiNER: disagreement-guided instruction refinement via simulating pilot annotation for zero-shot named entity recognition.**

Siun Kim, Hyung-Jin Yoon. *Proceedings of ACL 2026 (Main Conference)*.

2. **Questioning our questions: how well do medical QA benchmarks evaluate clinical capabilities of language models?**  
Siun Kim, Hyung-Jin Yoon. *24th Workshop on Biomedical Language Processing (BioNLP @ ACL 2025)* — Best Paper.
3. **Automatic extraction of comprehensive drug safety information from adverse drug event narratives in the Korea Adverse Event Reporting System using natural language processing techniques.**  
Siun Kim, Taegwan Kang, Tae Kyu Chung, Yoon Choi, YeSol Hong, Kyomin Jung, Howard Lee. *Drug Safety*, 2023.
4. **Beyond fine-tuning: leveraging domain-aware in-context learning with large language models for clinical named entity recognition.**  
Siun Kim, David Seung U Lee, Yujin Kim, Hyung-Jin Yoon, Howard Lee. *Journal of Biomedical Informatics*, 2026.
5. **CRese: benchmark data and automatic evaluation framework for recommending eligibility criteria from clinical trial information.**  
Siun Kim, Jung-Hyun Won, David Seung U Lee, Renqian Luo, Lijun Wu, Tao Qin, Howard Lee. *Findings of EACL 2024*.
6. **Revisiting machine learning based drug repurposing: drug indications are not a right prediction target.**  
Siun Kim, Jung-Hyun Won, David Seung U Lee, Renqian Luo, Lijun Wu, Yingce Xia, Tao Qin, Howard Lee. *Conference on Health, Inference, and Learning (CHIL 2023)*, PMLR.
7. **Intra-session recurrence in intradialytic hypotension prediction: evaluation implications and recurrence-aware modeling.**  
Siun Kim, Jiwon Ryu, Sejoong Kim, Su Hwan Kim, Myeongju Kim, Hyung-Jin Yoon. *Frontiers in Digital Health*, 2026.
8. **An annotated corpus from biomedical articles to construct a drug-food interaction database.**  
Siun Kim, Yoon Choi, Jung-Hyun Won, Jung Mi Oh, Howard Lee. *Journal of Biomedical Informatics*, 2022.
9. **A population pharmacokinetic-pharmacodynamic model of YH12852, a highly selective 5-hydroxytryptamine 4 receptor agonist, in healthy subjects and patients with functional constipation.**  
Siun Kim, Hyun A Lee, Seong Bok Jang, Howard Lee. *CPT: Pharmacometrics & Systems Pharmacology*, 2021.
10. **Large language model-assisted systematic review: validation based on Cochrane review data.**  
Siun Kim, Hyung-Jin Yoon. *Medical Informatics Europe (MIE) 2025*.

#### Other Publications

11. Improving clinical data standardization in surgical notes through biomedical entity linking with contextual evidence from large language models. Chaiho Shin et al. (Siun Kim, co-corresponding) *BMC Med. Inform. Decis. Mak.*, 2026.
12. One-year post-acute COVID-19 syndrome and mortality in South Korea: a nationwide matched cohort study using claims data. Jung-Hyun Won et al. *Front. Public Health*, 2024.
13. Deep learning-based long-term risk evaluation of incident type 2 diabetes using electrocardiogram in a non-diabetic population: a retrospective, multicentre study. Junmo Kim et al. *eClinicalMedicine*, 2024.
14. A critical review of the United States regulatory pathways for determining the equivalence of efficacy between CT-P13 and original infliximab (Remicade). Soohyun Kim<sup>†</sup>, Siun Kim<sup>†</sup> et al. (co-first) *Drug Des. Devel. Ther.*, 2020.

**Book:** *The era of biologics is coming*. Siun Kim, Howard Lee. (Selected as 2022 Sejong Book) Young Doc Press, 2019.

**Patent:** Apparatus and method for extracting drug-food interaction information. Howard Lee, Siun Kim, KR 10-2021-0172058 (application).

#### AWARDS & HONORS

**First Prize, Best Paper** — The 24th Workshop on Biomedical Language Processing (BioNLP @ ACL 2025), Vienna, Austria, Aug. 2025

**Best Article Presentation (1st)** — Korean Society for Clinical Pharmacology and Therapeutics, Nov. 2021

**Best Original Research Article (1st)** — The Korean Society of Medical Informatics, Oct. 2021

**Presidential Science Scholarship** — Korea Student Aid Foundation, 2013 – 2017

#### SKILLS & LANGUAGES

Python, TypeScript, SQL, R, PyTorch, JAX, scikit-learn, pandas, NumPy, SciPy, Transformers, BERTopic, LangChain, Anthropic API, OpenAI API, OpenRouter, Ollama, Langfuse, FAISS, Claude Code, FastAPI, React, Streamlit, PostgreSQL, SQLite, Git, GitHub Actions, pytest, Docker, Amazon S3, Amazon ECS, Amazon EC2, AWS Lambda, Amazon CloudWatch, Amazon Bedrock, NONMEM  
**Languages:** Korean (native), English (business fluent)